Ch 2: Polynomials

Question Bank | 2M | 3M | 5M

SECTION A - 2 MARKS

Q1. The graph of y = p(x) is given where the curve intersects the x-axis at 2 points. How many zeroes does p(x) have? What is the minimum degree of the polynomial?

Q2. The graph of y = p(x) touches the x-axis at exactly one point. How many zeroes does p(x) have?

Q3. The graph of y = p(x) does not intersect the x-axis at all. How many zeroes does p(x) have?

Q4. The graph of y = p(x) intersects the x-axis at 3 distinct points. What is the minimum degree of p(x)?

Q5. Find the zeroes of the polynomial $x^2 - 3$. ***(PYQ 2024)***

Q6. Find the zeroes of the polynomial $x^2 - 15$. ***(PYQ 2024)***

Q7. Find the zeroes of $4x^2 - 4x - 3$.

Q8. Find the zeroes of $6x^2 - 3 - 7x$.

Q9. Find the zeroes of $x^2 - 2x - 8$.

Q10. Find the zeroes of $2x^2 - x - 6$.

Q11. Find the zeroes of $3x^2 - x - 4$.

Q12. Find the zeroes of $\sqrt{3}\, x^2 + 10x + 7\sqrt{3}$.

Q13. Find the zeroes of $4x^2 + 5\sqrt{2}\, x - 3$.

Q14. Find the zeroes of $t^2 - 15$.

Q15. If sum of zeroes of $3x^2 - kx + 6$ is 3, find k. ***(PYQ 2012)***

Q16. If one zero of polynomial $(k^2 + 4)x^2 + 13x + 4k$ is the reciprocal of the other, find k. ***(PYQ 2016)***

Q17. If 2 is a zero of $p(x) = x^2 + 3x + k$, find k.

Q18. If one zero of $3x^2 - 8x + 2k + 1$ is seven times the other, find k. ***(PYQ 2019)***

Q19. What should be added to the polynomial $x^2 - 5x + 4$ so that 3 is a zero of the resulting polynomial? ***(PYQ 2024)***

Q20. If α and β are zeroes of $x^2 - 4x + 3$, find $\alpha + \beta$ and $\alpha\beta$.

Q21. If α and β are zeroes of $2x^2 + 5x + k$ and $\alpha\beta = -1$, find k.

Q22. If one zero of $5x^2 + 13x + k$ is the reciprocal of the other, find k.

Q23. If α and β are zeroes of $p(x) = x^2 - (k+6)x + 2(2k-1)$, find k if $\alpha + \beta = \alpha\beta/2$.

Q24. Find a quadratic polynomial whose zeroes are 2 and -3.

Q25. Find a quadratic polynomial whose zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

Q26. Find a quadratic polynomial whose zeroes are 0 and $\sqrt{5}$.

Q27. Find a quadratic polynomial whose sum of zeroes is -3 and product is 2.

Q28. Find a quadratic polynomial whose sum and product of zeroes are $\sqrt{2}$ and $1/3$ respectively.

Q29. Find a quadratic polynomial whose sum of zeroes is 0 and product is -1.

Q30. On dividing $x^3 - 3x^2 + x + 2$ by polynomial $g(x)$, the quotient and remainder are $x - 2$ and $-2x + 4$ respectively. Find $g(x)$.

Q31. On dividing $x^3 - 8x^2 + 20x - 10$ by $g(x)$, the quotient is $x - 4$ and remainder is 6. Find $g(x)$.

Q32. If α and β are zeroes of $x^2 - p(x + 1) - c$, show that $(\alpha+1)(\beta+1) = 1 - c$.

Q33. If α and β are zeroes of $x^2 + 7x + 7$, find $(1/\alpha + 1/\beta) - \alpha\beta$.

Q34. Find the zeroes of $9t^2 - 6t + 1$ and verify the relationship between zeroes and coefficients.

SECTION B - 3 MARKS

Q1. Find the zeroes of $4x^2 - 4x - 8$ and verify the relationship between zeroes and coefficients. ***(PYQ 2020)***

Q2. Find the zeroes of $x^2 - 2\sqrt{3}\,x - 24$ and verify the relationship between zeroes and coefficients.

Q3. Find the zeroes of $6x^2 - 7x - 3$ and verify the relationship between zeroes and coefficients. ***(PYQ)***

Q4. Find the zeroes of $2x^2 + 7x + 3$ and verify the relationship between zeroes and coefficients.

Q5. Find the zeroes of $3x^2 - 2$ and verify the relationship between zeroes and coefficients. ***(PYQ)***

Q6. Find the zeroes of $x^2 + 7x + 10$ and verify the relationship between zeroes and coefficients.

Q7. Find the zeroes of $p(x) = 4x^2 - 4x - 15$ and verify the relationship.

Q8. Show that $1/2$ and $-3/2$ are the zeroes of $4x^2 + 4x - 3$ and verify the relationship. ***(PYQ)***

Q9. Verify that 2, 3, and $1/2$ are zeroes of the polynomial $2x^3 - 9x^2 + 13x - 6$.

Q10. Verify that 3, -1, and $-1/3$ are zeroes of $3x^3 - 5x^2 - 11x - 3$ and verify the relationship. ***(PYQ)***

Q11. Find a quadratic polynomial whose zeroes are $(3 + \sqrt{2})$ and $(3 - \sqrt{2})$.

Q12. Find a quadratic polynomial whose zeroes are $(2 + \sqrt{3})$ and $(2 - \sqrt{3})$.

Q13. Find a quadratic polynomial whose zeroes are $(5 + 2\sqrt{3})$ and $(5 - 2\sqrt{3})$.

Q14. Find a quadratic polynomial, sum of whose zeroes is $\sqrt{2}$ and product is $-3/2$. Also find its zeroes. ***(PYQ 2018)***

Q15. Find a quadratic polynomial whose sum and product of zeroes are $-3/2\sqrt{5}$ and $-1/2$ respectively. Also find its zeroes.

Q16. If α and β are zeroes of $x^2 - 2x - 8$, find a polynomial whose zeroes are 2α and 2β. ***(PYQ 2019)***

Q17. If α and β are zeroes of $x^2 + 5x + 6$, find a polynomial whose zeroes are $(\alpha+1)$ and $(\beta+1)$. ***(PYQ)***

Q18. If α and β are zeroes of $x^2 - 6x + k$, find k if $3\alpha + 2\beta = 20$. ***(PYQ 2019)***

Q19. If α and β are zeroes of $4x^2 + 4x + 1$, find $\alpha^2 + \beta^2$ and $\alpha^3 + \beta^3$.

Q20. If α and β are zeroes of $x^2 + 4x + 3$, form a polynomial whose zeroes are $1 + \beta/\alpha$ and $1 + \alpha/\beta$. ***(PYQ 2019)***

Q21. If α and β are zeroes of $2x^2 + 3x - 6$, find the value of $\alpha^2 + \beta^2$ and $1/\alpha + 1/\beta$.

Q22. If α and β are zeroes of $x^2 - ax - b$, find $\alpha^2 + \beta^2$. ***(PYQ)***

Q23. If α and β are zeroes of the polynomial $f(x) = x^2 - 5x + k$ such that $\alpha - \beta = 1$, find k. ***(PYQ 2018)***

Q24. If one zero of $2x^2 - 3x + p$ is 3, find the other zero and the value of p.

Q25. If sum of zeroes of $kx^2 + 2x + 3k$ equals their product, find k.

Q26. Divide $3x^3 + x^2 + 2x + 5$ by $1 + 2x + x^2$ and verify division algorithm. ***(PYQ)***

Q27. Divide $x^3 - 3x^2 + 5x - 3$ by $x^2 - 2$ and verify division algorithm.

Q28. Divide $p(x) = x^4 - 5x + 6$ by $g(x) = 2 - x^2$ and find quotient and remainder.

Q29. Divide $3x^2 - x^3 - 3x + 5$ by $x - 1 - x^2$ and verify division algorithm. ***(PYQ)***

Q30. On dividing $3x^3 + 4x^2 + 5x - 13$ by g(x), the quotient and remainder were $3x + 10$ and $16 - 3x$ respectively. Find g(x). ***(PYQ)***

Q31. What must be subtracted from $8x^4 + 14x^3 - 2x^2 + 7x - 8$ so that the result is exactly divisible by $4x^2 + 3x - 2$? ***(PYQ HOTS)***

Q32. What must be added to $4x^4 + 2x^3 - 2x^2 + x - 1$ so that the result is divisible by $x^2 + 2x - 3$? ***(PYQ HOTS)***

Q33. Find all zeroes of $x^3 + 3x^2 - 2x - 6$ if one of its zeroes is $\sqrt{2}$.

Q34. Find all zeroes of $2x^3 - 4x - x^2 + 2$ if one zero is $\sqrt{2}$.

Q35. Find all zeroes of $x^3 - 3\sqrt{5}\, x^2 - 5x + 15\sqrt{5}$ if one zero is $\sqrt{5}$. ***(PYQ)***

Q36. If the polynomial $x^4 - 6x^3 + 16x^2 - 25x + 10$ is divided by $x^2 - 2x + k$ and the remainder is $x + a$, find k and a. ***(PYQ 2018)***

Q37. If α and β are zeroes of $x^2 - 3x + 2$, find a polynomial with zeroes $(1/2\alpha + \beta)$ and $(1/2\beta + \alpha)$.

Q38. Obtain all other zeroes of $x^4 - 7x^2 + 12$ given two zeroes are $\sqrt{3}$ and $-\sqrt{3}$.

Q39. If two zeroes of $x^4 - 6x^3 - 26x^2 + 138x - 35$ are $2 \pm \sqrt{3}$, find the other two zeroes. ***(PYQ 2018)***

Q40. Find all zeroes of $2x^4 - 9x^3 + 5x^2 + 3x - 1$ if two of its zeroes are $(2 + \sqrt{3})$ and $(2 - \sqrt{3})$. ***(PYQ 2019)***

Q41. Obtain all other zeroes of $3x^4 + 6x^3 - 2x^2 - 10x - 5$ if two of its zeroes are $\sqrt{(5/3)}$ and $-\sqrt{(5/3)}$. ***(PYQ 2019)***

Q42. If the zeroes of the cubic polynomial $x^3 - 6x^2 + 3x + 10$ are of the form a, a+b, and a+2b, find a and b. ***(PYQ)***

<u>SECTION C - 5 MARKS</u>

Q1. Find all zeroes of $2x^4 - 3x^3 - 3x^2 + 6x - 2$ if two zeroes are $\sqrt{2}$ and $-\sqrt{2}$. ***(PYQ 2022)***

Q2. Find all zeroes of $x^4 - 3\sqrt{2}\, x^3 - 3x^2 + 3\sqrt{2}\, x - 4$ if two zeroes are $\sqrt{2}$ and $2\sqrt{2}$. ***(PYQ 2020)***

Q3. Find all zeroes of $2x^4 - 9x^3 + 5x^2 + 3x - 1$ if two zeroes are $(2 + \sqrt{3})$ and $(2 - \sqrt{3})$. ***(PYQ 2019)***

Q4. If two zeroes of $x^4 - 6x^3 - 26x^2 + 138x - 35$ are $2 + \sqrt{3}$ and $2 - \sqrt{3}$, find the other two zeroes. ***(PYQ 2018)***

Q5. Find all zeroes of $3x^4 + 6x^3 - 2x^2 - 10x - 5$ if two zeroes are $\sqrt{(5/3)}$ and $-\sqrt{(5/3)}$. ***(PYQ 2019)***

Q6. Obtain all other zeroes of $x^4 + 4x^3 - 2x^2 - 20x - 15$ if two zeroes are $\sqrt{5}$ and $-\sqrt{5}$. ***(PYQ)***

Q7. Find all zeroes of $x^4 - 17x^2 + 16$ given two zeroes are 4 and 1.

Q8. Find all zeroes of $x^4 - 7x^2 + 12$ given two zeroes are $\sqrt{3}$ and $-\sqrt{3}$.

Q9. Find all zeroes of $6x^4 - 5x^3 - 19x^2 + 16x - 4$ if two zeroes are 2 and -2.

Q10. If the polynomial $x^4 - 6x^3 + 16x^2 - 25x + 10$ is divided by $x^2 - 2x + k$, and remainder is $x + a$, find k and a. Hence find all zeroes of the polynomial. ***(PYQ 2018)***

Q11. If α and β are zeroes of $x^2 + 4x + 3$, form polynomial whose zeroes are $1 + \beta/\alpha$ and $1 + \alpha/\beta$. Verify using division algorithm. ***(PYQ 2019)***

Q12. Divide $2x^4 - 5x^3 + x^2 + 3x - 2$ by $x^2 - 3x + 2$. Verify the division algorithm. Find the quotient and remainder. Are the zeroes of the divisor also zeroes of the dividend?

Q13. If α and β are zeroes of $2x^2 - 5x + 7$, find a polynomial whose zeroes are $(2\alpha + 3\beta)$ and $(3\alpha + 2\beta)$. ***(PYQ HOTS)***

Q14. Find all zeroes of $x^3 + 3x^2 - 2x - 6$ and verify using sum, sum of products in pairs, and product of all zeroes relationship with coefficients.

Q15. Find all zeroes of $2x^3 + x^2 - 6x - 3$ given that one zero is $-1/2$. Verify all three relationships between zeroes and coefficients of a cubic polynomial.

Q16. A polynomial $q(x) = x^3 - 2x^2 - 9x + k$, where k is a constant. The sum of two zeroes of q(x) is zero. Using the relationship between zeroes and coefficients, find: (i) zeroes of q(x) (ii) value of k. ***(PYQ 2025 expected)***

Q17. What must be subtracted from $x^4 + 2x^3 - 13x^2 - 12x + 21$ so that the result is exactly divisible by $x^2 - 4x + 3$? Also find the remaining zeroes after subtraction.

Q18. Find what must be added to the polynomial $4x^4 + 2x^3 - 2x^2 + x - 1$ so that it is exactly divisible by $x^2 + 2x - 3$. Hence find all zeroes of the resulting polynomial. ***(PYQ HOTS)***

Q19. If α and β are zeroes of $x^2 - 5x + k$, and $\alpha - \beta = 1$, find k. Hence form a polynomial whose zeroes are $(2\alpha - 1)$ and $(2\beta - 1)$. ***(PYQ 2018 extended)***

Q20. Verify that 2, 3 and 1/2 are zeroes of the polynomial $2x^3 - 9x^2 + 13x - 6$. Also verify all three relationships between zeroes and coefficients of a cubic polynomial. ***(PYQ)***

Q21. Find all zeroes of $x^4 - 8x^3 + 23x^2 - 28x + 12$ if two of its zeroes are 1 and 2.

Q22. If the zeroes of the cubic polynomial $x^3 - 6x^2 + 3x + 10$ are a, a+b, a+2b, find the values of a and b. Hence find all zeroes. ***(PYQ)***

Q23. Case Study: A student designs a rectangular garden whose area is represented by the polynomial $p(x) = 6x^2 + 11x - 10$. Find the zeroes of p(x) and verify the relationship. If one side of the garden is $(2x - 1)$, find the other side and verify using division algorithm.

Q24. Obtain all zeroes of $2x^4 + x^3 - 14x^2 - 19x - 6$ if two of its zeroes are −2 and −1.

Q25. Find all zeroes of $x^4 - 2x^3 - 7x^2 + 8x + 12$ if two zeroes are 3 and −2.